

CARLTON LINDSAY

Barcelona · open to remote, hybrid, or relocation

Design Engineer & Product Lead

> Designer who deploys. Engineer who dreams. I build what I design.

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Design Engineer taking AI-native interfaces from concept to production — knowledge-graph explorers, multi-agent pipeline UIs, and research instruments — owning both the interaction model and the production React/TypeScript behind it. A decade of hardware prototyping (Apple Vision Pro labs, VR hardware, medical devices) underneath the software. AI-first builder: Claude Code and Cursor are daily practice, not garnish.

CORE SKILLS

Build: React · TypeScript · Python · FastAPI · Node.js · GraphQL · SSE / real-time state in data-intensive apps · design systems & component libraries · data visualization & knowledge graphs

Design: interaction & motion design · information hierarchy for data-dense, trustworthy AI interfaces · design research (surveys, depth interviews) · rapid prototyping, software & hardware · accessibility, reduced motion, performance budgets

AI-native: Claude Code & Cursor agentic workflows as core practice · LLM integration & multi-agent UX · prompt-to-production pipelines · making agent reasoning legible to users

EXPERIENCE

Design Engineer & Product Lead — Futurity Systems, Barcelona Feb 2025 – Present

- ▶ **FAST — AI-accelerated science & foresight platform.** Lead Frontend Engineer & Lead UX/UI Designer: a data-dense research interface built for trust — GPU-accelerated knowledge-graph explorer that keeps ten-thousand-node graphs readable, React Flow analysis-workflow DAGs, multi-step LLM invention tools, plus a companion browser extension for one-action research capture. 400+ component production React/TypeScript codebase on a Chakra-based design system.
- ▶ **Futurity Engine — multi-agent research engine.** Built the UI concept prototype that established agent reasoning as a navigable graph, then the production frontend with the team: SSE streams every step of a five-phase agent pipeline into a live knowledge graph (25+ semantic node types) so the system stays legible while it works — AI earns trust when users can see what it's doing. Backend contributions in the FastAPI orchestrator.
- ▶ **Futurescaper — futures-exploration tool.** End-to-end ownership: solo rebuild from a colleague's browser-only proof-of-concept into a full product — UI, custom graph-layout system tuned for foresight data, and the backend owning AI orchestration.
- ▶ **Futures Garden — EU initiative (2025).** Concept and interaction design (team project): an exhibit where visitors talk to LLM-powered archetypes ("digital souls") by tapping NFC-tagged phygital objects on a physical Orb.
- ▶ **AI-native practice.** Claude Code and Cursor in the loop for scaffolding, exploration, and refactoring across all product work — including carlton.dev, designed and built end-to-end with AI agents as collaborators.

Hardware Associate Engineer — Apple (Vision Pro) 2022 – 2023

- ▶ **Lab buildout & operations.** Planned and built two prototyping labs for Vision Pro teams — floorplan, budget, equipment (3D printing, laser cutting, test gear): the Product Design team's on-site lab and the shared Hardware/Firmware lab. Ran them day to day alongside product-design, hardware, and firmware engineers; managed a small team; completed Apple management training.

Product Development Lab Manager — Pensar Development 2016 – 2022

- ▶ **Prototyping lab.** Ran the lab for a product-development consultancy — prototyping across consumer electronics, VR hardware, and medical devices (all under NDA): LabVIEW-automated test fixtures (PXI), CNC machining, SLS/FDM printing, wire harnessing, PCB rework.
- ▶ **Production leadership.** Planned and led a 500-unit hardware assembly program at a client factory: 8-person team, station design, training, and the test program.
- ▶ **The pivot to software.** Built a full-stack lab-equipment tracking portal with QR-code check-out (React, Node.js, GraphQL) — shipped to the whole engineering team.

Earlier — Senior PCB Repair Technician, FUJIFILM SonoSite (2015–16): component-level repair of ultrasound systems · Robotics Technician, Hydromax USA (2014–15): robotic inspection systems for water infrastructure.

EDUCATION & RESEARCH

Master's, Interaction Design — Harbour.Space University, Barcelona 2023 – 2025

- ▶ **Graduate research with Campus AI (2024).** Qualitative study of mid-career career-shifters for an LLM learning platform — 42-respondent survey, eight 90-minute depth interviews, 1,117 coded moments — synthesized into product strategy centered on capability growth over engagement (team of four).

PORTFOLIO

carlton.dev — live, working interactions, not screenshots: interactive case studies with honest ownership breakdowns. Near-zero framework JS, performance-budgeted, reduced-motion aware — typography, deliberate motion, considered interaction as craft.